EXISTENCE OF THE CENTRAL SERIES Z(U) OF THE GENERAL LINEAR SUPER YANGIAN WITH ITS TWO RTT-TYPE IDENTITIES

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Editorial note. The counted unit is Proposition 3.1 at source lines 877-886, source label zu, with the proof environment that begins on the next line at 888-977; the pairing is the ordinary adjacent one. The printed number is fixed by initial_counters so that every retained item keeps its original number: the source has the Introduction as section 1, so starting the counter at 1 makes the retained Definition of the Yangian section 2 (with its Proposition 2.1) and the Centre of the Yangian section 3, where the counted proposition prints as Proposition 3.1 exactly as in the published article.

Editorial note. The retained definition passages at source lines 169-351 and 368-422 are the author's own Section 1 on the definition of the Yangian: the graded tensor conventions, the generators and their defining relations, the permutation operator P, the R-matrix with (3.52), the series T(u) and T_p(u), Propositions 2.1 and 2.2 with their proofs, the antiautomorphism tau of End C^{M|N} defined at line 368, Proposition 2.3 with its proof, and the series T^S(u) at line 414. The counted proof uses tau in its first line through the operator Q, so both passages are retained; they carry the section heading of the source and are context, not a second counted proposition.

Editorial note. The context span at 863-876 is the opening of the Centre of the Yangian section, including the author's remark that the description of the centre comes from computing the square of the antipodal mapping. No retained passage was altered.

Editorial note. The retained preamble reproduces the author's macro and theorem declaration block verbatim from source lines 9-71, including the same theorem environment names the SIGMA publisher class declares; the publisher class itself is not installed on the assembly host, so the wrapper class amsart is used and the equation counter is still tied to the section counter as in the source.

Editorial note. The retained bibliography carries exactly the one work cited by the retained text (Nazarov, Quantum Berezinian and the classical Capelli identity), copied verbatim from the source's own reference list.

Editorial scope. One original result: Proposition 3.1 (label zu) of the source with its complete author proof at source lines 888-977, together with the author's own definition of the super Yangian that the proof uses. The already-counted Theorem 2.8 of the same source, the free-generation theorem of the same section and the quantum Berezinian shift theorem are not counted here and are not used as premises.

Allowed standard prerequisites (editorial).

- Super Yangian and its RTT presentation. For the general linear Lie superalgebra gl_{M|N} over C the Yangian Y(gl_{M|N}) is the associative

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Z_2-graded algebra with generators T_ij^(r) for r at least 1, whose generating series T_ij(u) satisfy the RTT relation; the inverse matrix series is written T'_ij(u) and the antipode is written S. The author sets all of this up at source lines 169-320, which are retained in the material, so nothing in the counted proof depends on an outside convention.. Fixes the algebra, its generators, the series T_ij(u) and T'_ij(u) and the antiautomorphism tau from which the operator Q is built.

- Formal power series in u^{-1} and the rational R-matrix. All series are formal in the variable u^{-1} with coefficients in the Yangian, and the rational R-matrix R(u) together with its twisted inverse R-hat_S(u) = 1 + Q(u-M+N)^{-1} is used through the identities recorded at source lines 249-308, which are retained in the material.. Supplies the identities (3.0), (3.3), (3.52) and (pop) that the counted proof quotes when it forms Q, R-hat_S and the three-leg computation.

2. Definition of the Yangian

Throughout this article we will use the following general conventions. Let A and B be any two associative $\mathbb{Z}_2$-graded algebras. Their tensor product $\mathrm{A}\otimes\mathrm{B}$ is also an associative $\mathbb{Z}_2$-graded algebra such that for any homogeneous elements $X, X' \in \mathrm{A}$ and $Y, Y' \in \mathrm{B}$

$$(X\otimes Y)(X'\otimes Y') = XX'\otimes YY'(-1)^{\deg X'\deg Y}, \tag{2.1}$$

$$\deg(X\otimes Y) = \deg X + \deg Y. \tag{2.2}$$

For any $\mathbb{Z}_2$-graded modules U and V over A and B respectively, the vector space $U\otimes V$ is a $\mathbb{Z}_2$-graded module over $\mathrm{A}\otimes\mathrm{B}$ such that for any homogeneous elements $x\in U$ and $y\in V$

$$(X\otimes Y)(x\otimes y) = Xx\otimes Yy(-1)^{\deg x\deg Y}, \tag{2.3}$$

$$\deg(x\otimes y) = \deg x + \deg y. \tag{2.4}$$

A homomorphism $\alpha\colon \mathrm{A}\to\mathrm{B}$ is a linear map such that $\alpha(XX') = \alpha(X)\alpha(X')$ for all $X, X'\in\mathrm{A}$. But an antihomomorphism $\beta\colon \mathrm{A}\to\mathrm{B}$ is a linear map such that for all homogeneous $X, X'\in\mathrm{A}$

$$\beta(XX') = \beta(X')\beta(X)(-1)^{\deg X\deg X'}. \tag{2.5}$$

If A is unital, let ι_h be its embedding into the tensor product $\mathrm{A}^{\otimes n}$ as the h-th tensor factor:

$$\iota_h(X) = 1^{\otimes(h-1)}\otimes X\otimes 1^{\otimes(n-h)} \qquad \text{for} \quad h = 1,\dots,n.$$

Here n can be any positive integer. We will also use various embeddings of the algebra $\mathrm{A}^{\otimes m}$ into $\mathrm{A}^{\otimes n}$ for any $m = 1,\dots,n$. For any choice of pairwise distinct indices $h_1,\dots,h_m \in \{1,\dots,n\}$ and of an element $X\in\mathrm{A}^{\otimes m}$ of the form $X = X^{(1)}\otimes\cdots\otimes X^{(m)}$ we will denote

$$X_{h_1\dots h_m} = \iota_{h_1}\big(X^{(1)}\big)\cdots\iota_{h_m}\big(X^{(m)}\big)\in\mathrm{A}^{\otimes n}.$$

We will then extend the notation $X_{h_1\dots h_m}$ to all elements $X\in\mathrm{A}^{\otimes m}$ by linearity.

Now let the indices i, j run through $1,\dots,M+N$. We will always write $\bar\imath = 0$ if $1\leqslant i\leqslant M$ and $\bar\imath = 1$ if $M < i\leqslant M+N$. Consider the $\mathbb{Z}_2$-graded vector space $\mathbb{C}^{M|N}$. Let $e_i\in\mathbb{C}^{M|N}$ be an element of the standard basis. The $\mathbb{Z}_2$-grading on $\mathbb{C}^{M|N}$ is defined so that $\deg e_i = \bar\imath$. Let $E_{ij}\in\operatorname{End}\mathbb{C}^{M|N}$ be the standard matrix unit, so that $E_{ij}e_k = \delta_{jk}e_i$. The associative algebra $\operatorname{End}\mathbb{C}^{M|N}$ is $\mathbb{Z}_2$-graded so that $\deg E_{ij} = \bar\imath + \bar\jmath$.

For any n we can identify the tensor product $\big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes n}$ with the algebra $\mathrm{End}\big(\big(\mathbb{C}^{M|N}\big)^{\otimes n}\big)$ acting on the vector space $\big(\mathbb{C}^{M|N}\big)^{\otimes n}$ by repeatedly using the conventions (2.3) and (2.4).

Let us introduce the *Yangian* of the Lie superalgebra $\mathfrak{gl}_{M|N}$. This is the complex associative unital $\mathbb{Z}_2$-graded algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$ with the countable set of generators

$$T_{ij}^{(r)}, \qquad \text{where} \quad r = 1, 2, \dots \quad \text{and} \quad i, j = 1, \dots, M+N. \tag{2.6}$$

The $\mathbb{Z}_2$-grading on the algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$ is determined by setting $\deg T_{ij}^{(r)} = \bar{\imath} + \bar{\jmath}$ for $r \geqslant 1$. To write down defining relations for these generators we will employ the series

$$T_{ij}(u) = \delta_{ij} \cdot 1 + T_{ij}^{(1)} u^{-1} + T_{ij}^{(2)} u^{-2} + \cdots \tag{2.7}$$

in a formal variable u with coefficients from $\mathrm{Y}(\mathfrak{gl}_{M|N})$. Then for all possible indices i, j, k, l

$$(u-v)[T_{ij}(u), T_{kl}(v)](-1)^{\bar{\imath}\bar{k}+\bar{\imath}\bar{l}+\bar{k}\bar{l}} = T_{kj}(u)T_{il}(v) - T_{kj}(v)T_{il}(u), \tag{2.8}$$

where v is another formal variable. The square brackets here stand for the supercommutator. Notice that the series denoted by $T_{ij}(u)$ here and in [1] differ by the scalar factor $(-1)^{(\bar{\imath}+1)\bar{\jmath}}$.

We will also use the following matrix form of the defining relations (2.8). Take the element

$$P = \sum_{i,j=1}^{M+N} E_{ij} \otimes E_{ji}(-1)^{\bar{\jmath}} \in \big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes 2}. \tag{2.9}$$

This element acts on the vector space $\big(\mathbb{C}^{M|N}\big)^{\otimes 2}$ so that $e_i \otimes e_j \mapsto e_j \otimes e_i(-1)^{\bar{\imath}\bar{\jmath}}$. Here we identify the algebra $\big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes 2}$ with the algebra $\mathrm{End}\,\big(\big(\mathbb{C}^{M|N}\big)^{\otimes 2}\big)$ by using (2.3).

For any n let $\mathfrak{S}_n$ be the symmetric group acting on the set $\{1, \dots, n\}$ by permutations. For each $m = 1, \dots, n-1$ denote by σ_m the element of $\mathfrak{S}_n$ exchanging m and $m+1$. The group $\mathfrak{S}_n$ also acts on the vector space $\big(\mathbb{C}^{M|N}\big)^{\otimes n}$. This action is defined by the assignment $\sigma_m \mapsto P_{m,m+1}$ for each m. Here we identify the algebra $\big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes n}$ with $\mathrm{End}\,\big(\big(\mathbb{C}^{M|N}\big)^{\otimes n}\big)$ via (2.3), (2.4).

The rational function $R(u) = 1 - Pu^{-1}$ with values in the algebra $\big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes 2}$ is called the *Yang R-matrix*. It satisfies the *Yang–Baxter equation* in the algebra $\big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes 3}(u, v, w)$

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v). \tag{2.10}$$

Since $P^2 = 1$, we also have the relation

$$R(-u)R(u) = 1 - u^{-2}. \tag{2.11}$$

Now combine all the series (2.7) into the single element

$$T(u) = \sum_{i,j=1}^{M+N} E_{ij} \otimes T_{ij}(u) \in \big(\mathrm{End}\,\mathbb{C}^{M|N}\big) \otimes \mathrm{Y}(\mathfrak{gl}_{M|N})\big[\big[u^{-1}\big]\big]. \tag{2.12}$$

For any n and any $p = 1, \dots, n$ we will denote

$$T_p(u) = \iota_p \otimes \mathrm{id}(T(u)) \in \big(\mathrm{End}\,\mathbb{C}^{M|N}\big)^{\otimes n} \otimes \mathrm{Y}(\mathfrak{gl}_{M|N})\big[\big[u^{-1}\big]\big]. \tag{2.13}$$

By using this notation for $n = 2$ the relations (2.8) can be rewritten as

$$(R(u-v) \otimes 1)T_1(u)T_2(v) = T_2(v)T_1(u)(R(u-v) \otimes 1). \tag{2.14}$$

Namely, after multiplying each side of (2.14) by $u-v$ it becomes a relation of series in u, v with coefficients in $\big(\operatorname{End}\mathbb{C}^{M|N}\big)^{\otimes 2}\otimes \mathrm{Y}(\mathfrak{gl}_{M|N})$ equivalent to the collection of all the relations (2.8).

Proposition 2.1. *An antiautomorphism of* $\mathrm{Y}(\mathfrak{gl}_{M|N})$ *can be defined by the assignment*

$$T_{ij}(u)\mapsto T_{ij}(-u). \tag{2.15}$$

Proof. Due to the convention (2.1), by using the notation (2.13) for $n=2$ we get

$$T_1(u)T_2(v)=\sum_{i,j,k,l=1}^{M+N} E_{ij}\otimes E_{kl}\otimes T_{ij}(u)T_{kl}(v)(-1)^{(\bar\imath+\bar\jmath)(\bar k+\bar l)}, \tag{2.16}$$

$$T_2(-v)T_1(-u)=\sum_{i,j,k,l=1}^{M+N} E_{ij}\otimes E_{kl}\otimes T_{kl}(-v)T_{ij}(-u). \tag{2.17}$$

By using the convention (2.5), the antihomomorphism property of (2.15) follows from the relation

$$(R(u-v)\otimes 1)T_2(-v)T_1(-u)=T_1(-u)T_2(-v)(R(u-v)\otimes 1),$$

which is obtained from (2.14) by using (2.11). The antihomomorphism (2.15) is clearly involutive and therefore bijective. □

For all indices i, j define the series $T'_{ij}(u)$ by using the element inverse to (2.12) so that

$$T(u)^{-1}=\sum_{i,j=1}^{M+N} E_{ij}\otimes T'_{ij}(u). \tag{2.18}$$

Proposition 2.2. *An antiautomorphism of* $\mathrm{Y}(\mathfrak{gl}_{M|N})$ *can be defined by the assignment*

$$T_{ij}(u)\mapsto T'_{ij}(u). \tag{2.19}$$

Further, let τ be the antiautomorphism of $\operatorname{End}\mathbb{C}^{M|N}$ defined by the assignment

$$E_{ij}\mapsto E_{ji}(-1)^{\bar\imath(\bar\jmath+1)}.$$

Then by the definition (2.12) we have

$$\tau\otimes\mathrm{id}(T(u))=\sum_{i,j=1}^{M+N} E_{ji}\otimes T_{ij}(u)(-1)^{\bar\imath(\bar\jmath+1)}=\sum_{i,j=1}^{M+N} E_{ij}\otimes T_{ji}(u)(-1)^{\bar\jmath(\bar\imath+1)}.$$

Proposition 2.3. *An antiautomorphism of* $\mathrm{Y}(\mathfrak{gl}_{M|N})$ *can be defined by the assignment*

$$T_{ij}(u)\mapsto T_{ji}(u)(-1)^{\bar\jmath(\bar\imath+1)}. \tag{2.20}$$

Proof. Observe that $(\tau\otimes\tau)(P)=P$ and hence

$$(\tau\otimes\tau)(R(u-v))=R(u-v). \tag{2.21}$$

Therefore the antihomomorphism property of (2.20) follows from the relation which is obtained by applying $\tau\otimes\tau\otimes\mathrm{id}$ to both sides of(2.14), see the proofs of Propositions 2.1 and 2.2 above. To prove the bijectivity of the antihomomorphism (2.20), observe that its square is given by

$$T_{ij}(u)\mapsto T_{ij}(u)(-1)^{\bar\imath+\bar\jmath}.$$

In particular, the square is an automorphism of the $\mathbb{Z}_2$-graded algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$. □

Put $T^{\sharp}(u) = \tau \otimes \mathrm{id}\big(T(u)^{-1}\big)$. Then by (2.18)

$$T^{\sharp}(u) = \sum_{i,j=1}^{M+N} E_{ji} \otimes T'_{ij}(u)(-1)^{\bar{\imath}(\bar{\jmath}+1)} = \sum_{i,j=1}^{M+N} E_{ij} \otimes T'_{ji}(u)(-1)^{\bar{\jmath}(\bar{\imath}+1)}. \tag{2.22}$$

3. Centre of the Yangian

Here we will give a description of the centre of the algebra $\mathrm{Y}(\mathfrak{gl}_{M|N})$. An element of $\mathrm{Y}(\mathfrak{gl}_{M|N})$ is called *central* if it supercommutes with each element of $\mathrm{Y}(\mathfrak{gl}_{M|N})$. However, we will see that the central elements of $\mathrm{Y}(\mathfrak{gl}_{M|N})$ have the $\mathbb{Z}_2$-degree 0. Hence they commute with each element of $\mathrm{Y}(\mathfrak{gl}_{M|N})$ in the usual sense. Our description comes from computing the square of the antipodal mapping S of $\mathrm{Y}(\mathfrak{gl}_{M|N})$. Another description of the centre of $\mathrm{Y}(\mathfrak{gl}_{M|N})$ will be given in the next section. In that section we will also establish a correspondence between the two descriptions.

Proposition 3.1. *There is a formal power series $Z(u)$ in u^{-1} with coefficients in the centre of $\mathrm{Y}(\mathfrak{gl}_{M|N})$ and with leading term 1 such that for all indices i and j*

$$\sum_{k=1}^{M+N} T_{kj}(u+M-N)T'_{ik}(u) = \delta_{ij}Z(u), \tag{3.1}$$

$$\sum_{k=1}^{M+N} T'_{kj}(u)T_{ik}(u+M-N) = \delta_{ij}Z(u). \tag{3.2}$$

Proof. By using the definition (2.9) introduce the element of the algebra $\big(\operatorname{End}\mathbb{C}^{M|N}\big)^{\otimes 2}$

$$Q = (\mathrm{id}\otimes\tau)(P) = \sum_{i,j=1}^{M+N} E_{ij} \otimes E_{ij}(-1)^{\bar{\imath}\bar{\jmath}}. \tag{3.3}$$

The image of the action of Q on $\big(\mathbb{C}^{M|N}\big)^{\otimes 2}$ is one dimensional and is spanned by the vector

$$\sum_{i=1}^{M+N} e_i \otimes e_i.$$

Here we regard Q as an element of the algebra $\operatorname{End}\big(\big(\mathbb{C}^{M|N}\big)^{\otimes 2}\big)$ by identifying this algebra with $\big(\operatorname{End}\mathbb{C}^{M|N}\big)^{\otimes 2}$ via (2.3). We also have $Q^2 = (M-N)Q$. By using the latter relation we get

$$((\mathrm{id}\otimes\tau)(R(u)))^{-1} = \big(1 - Qu^{-1}\big)^{-1} = 1 + Q(u-M+N)^{-1}. \tag{3.4}$$

The rational function of the variable u given by the equalities (3.4) will be denoted by $R^{\sharp}(u)$.

Now multiply both sides of the relation (2.14) by $T_2^{-1}(v)$ on the left and right, and then apply τ relative to the second tensor factor $\operatorname{End}\mathbb{C}^{M|N}$ in $\big(\operatorname{End}\mathbb{C}^{M|N}\big)^{\otimes 2}\otimes \mathrm{Y}(\mathfrak{gl}_{M|N})$. Multiplying the resulting relation on the left and right by $R^{\sharp}(u-v)\otimes 1$ yields

$$\big(R^{\sharp}(u-v)\otimes 1\big)T_1(u)T_2^{\sharp}(v) = T_2^{\sharp}(v)T_1(u)(R^{\sharp}(u-v)\otimes 1). \tag{3.5}$$

Multiplying the latter relation by $u-v-M+N$ and then setting $u = v+M-N$ we get

$$(Q\otimes 1)T_1(v+M-N)T_2^{\sharp}(v) = T_2^{\sharp}(v)T_1(v+M-N)(Q\otimes 1), \tag{3.6}$$

see (3.4). As the image of Q in $\big(\mathbb{C}^{M|N}\big)^{\otimes 2}$ is one dimensional, either side of the relation (3.6) must be equal to $Q\otimes Z(v)$ where $Z(v)$ is a certain power series in v^{-1}

with coefficients from $\mathrm{Y}(\mathfrak{gl}_{M|N})$. The equality of the left-hand side and of the right-hand side of (3.6) to $Q\otimes Z(v)$ is equivalent respectively to (3.1) and (3.2). We just need to replace the variable v in (3.6) by u.

It is immediate from (2.7) and (3.1) that every coefficient of the series $Z(u)$ has $\mathbb{Z}_2$-degree 0 and that its leading term is 1. Let us prove that all these coefficients are central in $\mathrm{Y}(\mathfrak{gl}_{M|N})$.

To this end we will work with the elements (2.13) where $n=3$. By using (2.14) and (3.5),

$$\begin{aligned}&\big(R^{\sharp}_{13}(u-v)R_{12}(u-v-M+N)\otimes 1\big)T_1(u)T_2(v+M-N)T^{\sharp}_3(v)\\&=\big(R^{\sharp}_{13}(u-v)\otimes 1\big)T_2(v+M-N)T_1(u)T^{\sharp}_3(v)(R_{12}(u-v-M+N)\otimes 1)\\&=T_2(v+M-N)T^{\sharp}_3(v)T_1(u)\big(R^{\sharp}_{13}(u-v)R_{12}(u-v-M+N)\otimes 1\big).\end{aligned}\tag{3.7}$$

On the other hand, due to (2.11) we have the identity in the algebra $\big(\operatorname{End}\mathbb{C}^{M|N}\big)^{\otimes 3}(u)$

$$R_{13}(-u)P_{23}R_{12}(u)=\big(1-u^{-2}\big)P_{23}.$$

By applying to it the antiautomorphism τ relative to the third tensor factor of $\big(\operatorname{End}\mathbb{C}^{M|N}\big)^{\otimes 3}$ and then using the definition (3.4) we get

$$Q_{23}R^{\sharp}_{13}(u+M-N)R_{12}(u)=\big(1-u^{-2}\big)Q_{23}.\tag{3.8}$$

Therefore multiplying the first and third lines of the display (3.7) by $Q_{23}\otimes 1$ on the left yields

$$\begin{aligned}&\big(1-(u-v-M+N)^{-2}\big)T_1(u)(Q_{23}\otimes Z(v))\\&=(Q_{23}\otimes Z(v))T_1(u)\big(1-(u-v-M+N)^{-2}\big).\end{aligned}$$

We just need to replace the variable u in (3.8) by $u-v-M+N$ and use the relation (3.6). The last relation implies that any generator $T^{(r)}_{ij}$ commutes with every coefficient of $Z(v)$. □

References

[1] Nazarov M.L., Quantum Berezinian and the classical Capelli identity, *Lett. Math. Phys.* **21** (1991), 123–131.